VF Asia Footwear Laboratory Testing Manual

Test Method for Pull Strength of Sandals or Slipper Straps	**Test Method #: VF- F06** **Page 1 of 5**
Date Issued : May,2019 **Revision #: 2** **Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material.**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the pull strength of sandals or slipper straps.

REFERENCE: ASTM D903

EQUIPMENTS:

1) Tensile testing machine meeting the requirements of ASTM D 638.
2) Clamping jaws for the tensile tester
3) Testing jigs for toe fixed sandals (Thong-Type)
4) Razor or sharp knife or pair of scissors.

MATERIALS: Assembled shoes, sandals, slippers or upper components with eyelet or hook.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Following conditioning, check if cutting will be made base on the type of test to be conducted.

REPETITIONS: Test at least 1 pair

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

I. Thong Type / Y-Strap :

1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.

2. Fix the tool into the lower clamp of the tensile machine.

3. When pulling the toe post, remove the lateral and medial straps, then place the toe post at the center of the hole on the plate.

4. Fix the jig under, ensure the toe post is parallel from the clamps. (See figure 1)

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5. When pulling the side straps (lateral and medial straps), remove the toe post and one of the side straps. Align the hole with the strap.
6. Fix the jig under, ensure the strap is parallel from the clamps. (See figure 2)
7. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100mm/minute and the load range to 20kgf.
8. Start pulling test. Stop the test until the toe post or the strap is broken or separates from the foot-bed / sole

II. Vamp Type:

A. For vamp which is about 2 inches and below use jigs similar to thong side straps - see figure 2. Follow steps 1-8.

B. For vamp wider than 2 inches follow procedure below.

1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.
2. Insert the vamp to the shoe last type of jig then clamp to the upper clamp).
3. Clamp the outsole to the lower clamp of the tensile machine.
4. Ensure that the upper jig is parallel to the lower clamp. See figure 3.
5. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100mm/minute and the load range to 20kgf.
6. Start pulling test. Stop the test until the vamp is broken or separates from the foot-bed / sole.

TEST REPORT: highest peak @ kgf

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III. Back Strap Wrap Type:

Note: This is additional test conducted aside from pull strength test for Thong Type.

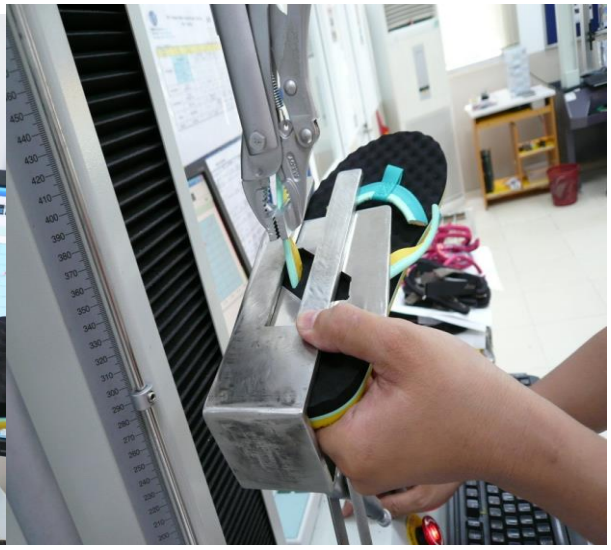
1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.
2. Clamp one side of wrap strap to the lower clamp and the other side to the upper clamp of the tensile machine.
3. Ensure that the 2 ends of wrap strap specimens are parallel. See figure 4.
4. Zero and calibrate the tensile testing machine .Set the crosshead speed to 100mm/minute and the load range to 20kgf.
5. Start pulling test. Stop the test until the back wrap strap is broken or the 2 ends separate from each other.

TEST REPORT: highest peak @ kgf

Figure 1



Figure 2

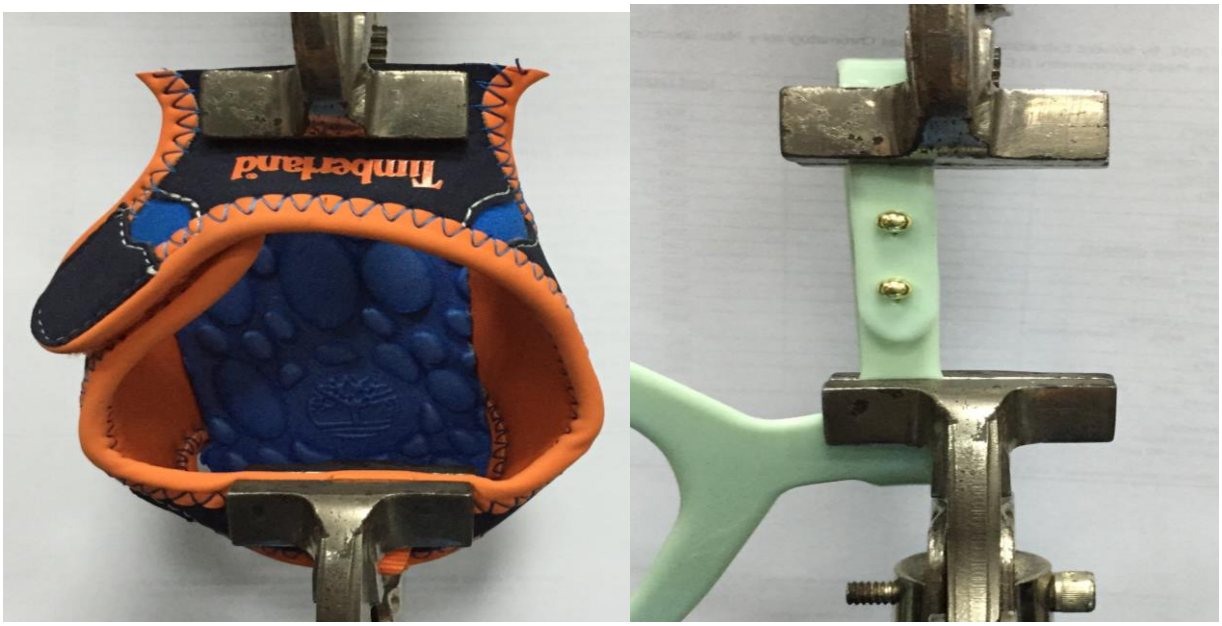


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Figure 3:



Figure 4 : Back Strap - with Velcro as closure



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Figure 4: Back Strap - with Rivets as closure :

